

5. Vectors		
5.1	The vector product $\mathbf{a} \times \mathbf{b}$ and the triple scalar product $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$.	The interpretation of $ \mathbf{a} \times \mathbf{b} $ as an area and $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ as a volume.
5.2	Use of vectors in problems involving points, lines and planes. The equation of a line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$.	Students may be required to use equivalent cartesian forms also. Applications to include (i) distance from a point to a plane, (ii) line of intersection of two planes, (iii) shortest distance between two skew lines.
5.3	The equation of a plane in the forms $\mathbf{r} \cdot \mathbf{n} = p$, $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$.	Students may be required to use equivalent cartesian forms also.